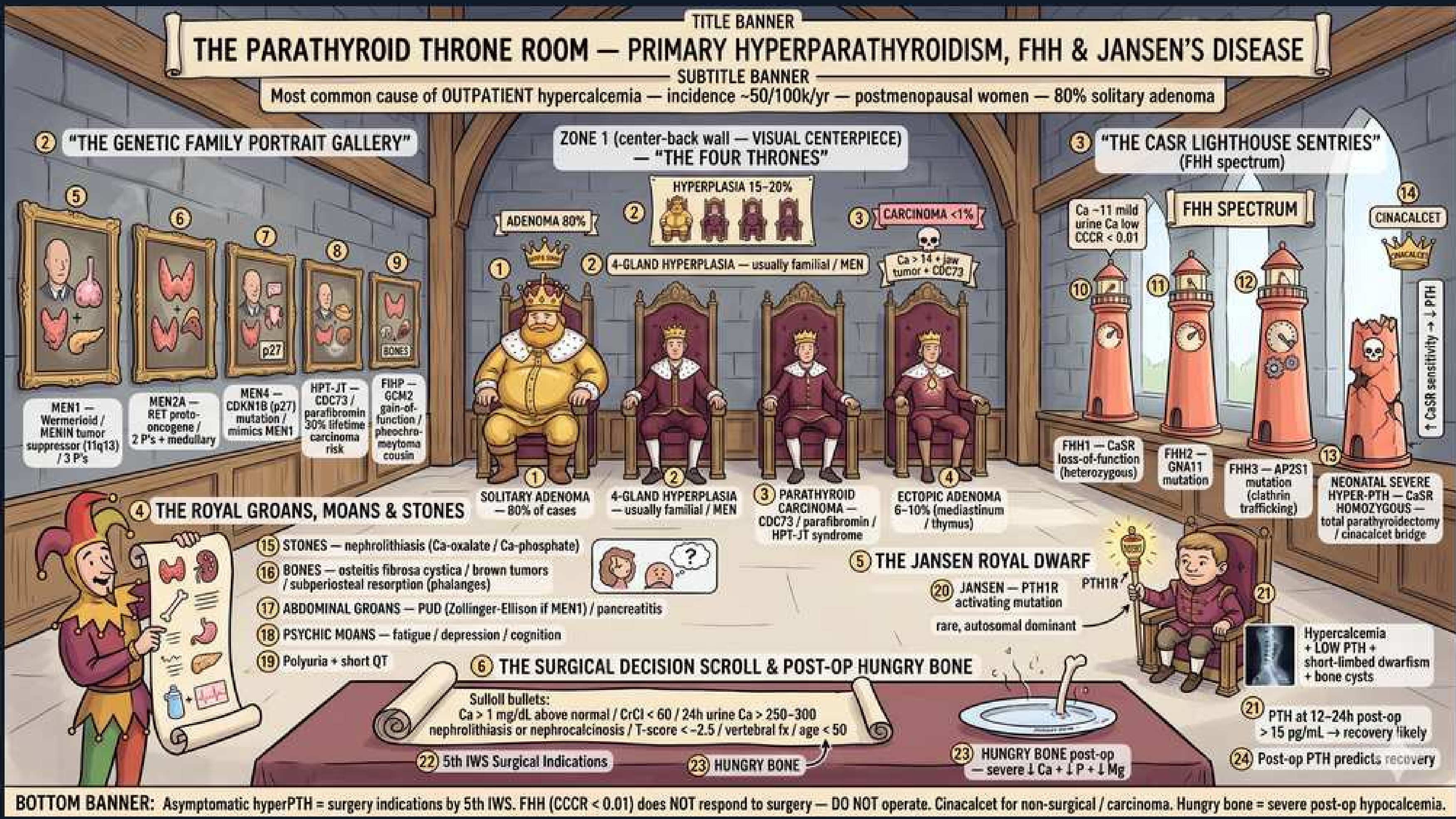
Primary Hyperparathyroidism, FHH & Jansen Disease



1. Solitary adenoma — 80% of cases

WHAT YOU SEE

Crowned king on center throne

WHAT IT ENCODES

A single parathyroid ADENOMA causes ~80% of primary hyperPTH — the most common cause of OUTPATIENT hypercalcemia (incidence ~50/100k/yr, peaks in postmenopausal women). High Ca with INAPPROPRIATELY high/normal PTH.

BOARD TRAP

Inpatient hypercalcemia is most often MALIGNANCY (PTHrP, lytic, 1,25). PTH HIGH = primary hyperPTH; PTH SUPPRESSED with high Ca = malignancy/other — PTH level splits them instantly.

2. 4-gland hyperplasia — MEN

WHAT YOU SEE

Four occupied thrones

WHAT IT ENCODES

Four-gland hyperplasia (~15%) is usually familial and tied to MEN syndromes. MEN1 (parathyroid + pituitary + pancreas) and MEN2A (parathyroid + medullary thyroid Ca + pheo).

BOARD TRAP

In MEN, ALL four glands are abnormal — subtotal (3.5-gland) parathyroidectomy is needed, not single-gland excision; a solitary adenoma resection would miss disease.

3. Parathyroid carcinoma <1%

WHAT YOU SEE

Dark ominous throne figure

WHAT IT ENCODES

Parathyroid carcinoma (<1%) causes very high Ca (often >14) and very high PTH, palpable neck mass, and is linked to CDC73/HRPT2 (HPT-jaw tumor syndrome).

BOARD TRAP

Markedly elevated Ca AND PTH with a palpable neck mass = think CARCINOMA — do NOT use cinacalcet as primary therapy; en bloc surgical resection is needed.

4. Ectopic adenoma (thymus)

WHAT YOU SEE

Misplaced gland icon

WHAT IT ENCODES

~6–10% of adenomas are ectopic (mediastinum/thymus, intrathyroidal) — a cause of failed/persistent disease after neck exploration. Sestamibi + 4D-CT localizes them.

BOARD TRAP

Persistent hypercalcemia after parathyroidectomy with an unexplored mediastinum = ectopic/supernumerary gland — re-image, don't assume cure failure of technique.

5. MEN1 — menin

WHAT YOU SEE

Genetic family portrait gallery

WHAT IT ENCODES

MEN1 (menin tumor suppressor, 11q13): parathyroid hyperplasia + pituitary adenoma + pancreatic NETs (gastrinoma/insulinoma). Hyperparathyroidism is usually the FIRST manifestation.

BOARD TRAP

Young patient with multigland hyperPTH + kidney stones + peptic ulcers (gastrinoma) = MEN1 — screen the family; a single-gland operation will recur.

6. MEN2A — RET

WHAT YOU SEE

RET-mutation family figure

WHAT IT ENCODES

MEN2A (RET proto-oncogene): medullary thyroid carcinoma + pheochromocytoma + parathyroid hyperplasia. Always rule out pheo BEFORE any neck/adrenal surgery.

BOARD TRAP

Operate on the PHEO first — unrecognized pheo during parathyroid/thyroid surgery can trigger fatal hypertensive crisis.

7. HPT-JT — CDC73 / parafibromin

WHAT YOU SEE

Jaw-tumor family figure

WHAT IT ENCODES

Hyperparathyroidism-jaw tumor syndrome (CDC73/HRPT2, parafibromin): parathyroid tumors (high carcinoma risk) + ossifying jaw fibromas + renal lesions.

BOARD TRAP

HyperPTH with a jaw tumor raises the parathyroid CARCINOMA probability sharply — CDC73 loss is shared with sporadic parathyroid cancer.

8. Stones — nephrolithiasis

WHAT YOU SEE

Jester scroll: stones

WHAT IT ENCODES

Primary hyperPTH classic 'stones, bones, groans, psychic moans': nephrolithiasis (Ca-oxalate/phosphate) and nephrocalcinosis from hypercalciuria.

BOARD TRAP

Symptomatic stones are a SURGICAL indication for parathyroidectomy even with only mild hypercalcemia.

9. Bones — osteitis fibrosa cystica

WHAT YOU SEE

Brown tumor / bone scroll

WHAT IT ENCODES

Chronic PTH excess → osteitis fibrosa cystica with subperiosteal resorption (radial phalanges), salt-and-pepper skull, and brown tumors (osteoclast-rich, hemorrhagic).

BOARD TRAP

A lytic 'brown tumor' can mimic a metastasis or giant-cell tumor radiologically — check Ca/PTH before biopsy; it resolves after parathyroidectomy.

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10. Abdominal groans — PUD/pancreatitis

WHAT YOU SEE

Scroll: GI groans

WHAT IT ENCODES

Hypercalcemia causes constipation, peptic ulcer disease (gastrin), and pancreatitis. Psychiatric 'moans': fatigue, depression, confusion.

BOARD TRAP

Hypercalcemia SHORTENS the QT interval — the inverse of hypocalcemia; a common ECG board swap.

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11. Polyuria — short QT

WHAT YOU SEE

ECG short QT, polyuria

WHAT IT ENCODES

Hypercalcemia impairs renal concentrating ability (nephrogenic DI) → polyuria/polydipsia and dehydration, and shortens the QT interval.

BOARD TRAP

Volume depletion worsens hypercalcemia — first-line acute therapy is IV saline; loop diuretics are NOT routine first-line (only after euvolemia).

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12. FHH — CaSR loss-of-function

WHAT YOU SEE

CaSR lighthouse sentries

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia (FHH): heterozygous CaSR LOSS-of-function → higher Ca set-point → mild lifelong hypercalcemia with HIGH/normal PTH but LOW urine Ca. Benign; no surgery.

BOARD TRAP

Calculate the Ca/Cr clearance ratio: FHH is <0.01, primary hyperPTH is >0.02. Missing FHH leads to a useless parathyroidectomy — always check urine Ca before operating.

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13. Neonatal severe hyperPTH (homozygous CaSR)

WHAT YOU SEE

Severe-mutation infant figure

WHAT IT ENCODES

Homozygous CaSR loss-of-function = neonatal severe hyperparathyroidism — life-threatening hypercalcemia in infancy requiring urgent total parathyroidectomy.

BOARD TRAP

Two FHH-carrier parents can produce a homozygous neonate with severe, surgical hypercalcemia — a different disease from benign heterozygous FHH.

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14. Jansen disease — PTH1R activating

WHAT YOU SEE

Dwarf figure, activated PTH1R

WHAT IT ENCODES

Jansen metaphyseal chondrodysplasia: CONSTITUTIVELY activating PTH1 receptor mutation → PTH-like effects WITHOUT high PTH: hypercalcemia, hypophosphatemia, and short-limbed dwarfism. Autosomal dominant.

BOARD TRAP

Hypercalcemia + hypophosphatemia with LOW PTH and skeletal dysplasia = receptor activation (Jansen), not an adenoma — PTH is low because the receptor is already 'on'.

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15. Cinacalcet — calcimimetic

WHAT YOU SEE

Cinacalcet pill panel

WHAT IT ENCODES

Cinacalcet (calcimimetic) sensitizes the CaSR → lowers PTH and Ca. Used for parathyroid carcinoma, severe primary hyperPTH not surgical candidates, and secondary/tertiary hyperPTH in CKD.

BOARD TRAP

Cinacalcet controls Ca but does NOT shrink the tumor or replace curative surgery in operable primary hyperPTH/carcinoma — it is a medical bridge, not a cure.

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16. Hungry bone post-op

WHAT YOU SEE

Post-op recovery / low Ca panel

WHAT IT ENCODES

After parathyroidectomy for severe hyperPTH, hungry bone syndrome causes prolonged hypocalcemia (+ low phosphate, low Mg) at ~12–24 h as demineralized bone reuptakes minerals; predicted by high pre-op ALP/PTH.

BOARD TRAP

Post-op hypocalcemia from HUNGRY BONE (low phosphate) differs from surgical hypoPTH (high phosphate) — check phosphate to tell them apart and replete aggressively.
